FinTech Risks and Challenges

A Case Study of HSBC Holdings plc

Part 1 – Written report research and drafting guide

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Prepared for independent study

1. Introduction and purpose

Financial technology (FinTech) refers to the use of digital technologies to deliver, support or improve financial products and services. Examples include mobile and online banking, application programming interfaces (APIs), cloud computing, artificial intelligence (AI), machine learning, blockchain, digital identity systems and real-time payment platforms. These technologies can make banking faster, more convenient and more efficient. They can also create new dependencies, vulnerabilities and forms of competition.

HSBC Holdings plc is an international banking and financial services group. As a large bank operating across multiple markets, HSBC must manage technology risks alongside credit, market, liquidity, conduct and compliance risks. FinTech matters because digital services are increasingly connected to customers, employees, suppliers, payment networks and other financial institutions. A disruption or security event in one part of this network can therefore have consequences beyond the original system.

Report purpose: This report examines (1) how FinTech creates new sources of risk and challenges for HSBC and (2) how participation in FinTech markets can increase risk within the firm. The central argument to develop is that FinTech is not inherently harmful; risk depends on how quickly it is adopted, how well it is governed, and whether controls keep pace with innovation.

2. How FinTech creates new sources of risk

2.1 Cybersecurity and expanding attack surfaces

Digital banking, mobile applications, APIs, cloud platforms and remote access tools create more points that criminals may attempt to exploit. Threats include phishing, credential theft, ransomware, malware, software vulnerabilities, denial-of-service attacks and attacks on suppliers. A successful attack could interrupt customer access, expose confidential information, enable fraudulent transactions or require costly recovery work. Interconnected systems can also allow an incident at one provider to affect several services. In your own analysis, distinguish prevention (secure design and access controls), detection (monitoring and threat intelligence) and response (containment, recovery and customer communication).

2.2 Data privacy and information governance

Digital services depend on collecting, storing, analysing and sharing customer and transaction data. This creates risks if data is collected without a clear purpose, retained for too long, accessed by unauthorised users, transferred insecurely or used in ways customers do not expect. Large-scale data incidents can trigger regulatory scrutiny, remediation costs, litigation and loss of trust. Cross-border banking adds complexity because privacy and data-localisation rules may differ between jurisdictions. Consider how HSBC should govern data quality, access permissions, retention, encryption and customer consent.

2.3 Artificial intelligence and model risk

AI and machine-learning tools may support fraud detection, customer service, credit decisions, compliance monitoring and operational processes. Their use creates model risk: outputs may be inaccurate, biased, unstable or difficult to explain. A model trained on incomplete or historically biased data may produce unfair outcomes. Generative AI can also produce plausible but incorrect information, expose confidential data through inappropriate prompts, or create new opportunities for impersonation and fraud. Human oversight, testing, documentation, explainability, independent validation and ongoing performance monitoring are therefore important.

2.4 Operational resilience and service continuity

Banks rely on technology to process payments, maintain account records, authenticate customers and support front-office and back-office activities. Software defects, failed upgrades, capacity constraints, network problems or cloud outages may interrupt essential services. Even when no data is stolen, customers may be unable to access funds or complete urgent transactions. Operational resilience requires identifying important business services,

mapping their technology dependencies, setting impact tolerances, testing recovery arrangements and learning from incidents.

2.5 Third-party, cloud and concentration risk

FinTech participation often involves specialist vendors, cloud providers, data processors, software companies and payment partners. Outsourcing can provide expertise and economies of scale, but it does not remove the bank's responsibility to manage risk. A provider may suffer a cyberattack, become insolvent, change its service, fail to meet regulatory expectations or experience an outage. Dependence on a small number of major providers can create concentration risk: several services may be affected by one common failure. Contracts, due diligence, audit rights, exit plans and tested alternatives are important controls.

2.6 Financial crime, fraud and identity risks

Digital onboarding and fast payments can improve convenience but may also be abused through synthetic identities, account takeover, social engineering, mule accounts and automated scams. Deepfake audio or video can make impersonation more convincing. Faster payment systems may reduce the time available to identify and stop suspicious transfers. Banks need risk-based customer due diligence, transaction monitoring, sanctions screening, strong authentication and procedures for investigating suspicious activity.

2.7 Regulatory, legal and conduct challenges

Financial institutions must comply with requirements covering data protection, cybersecurity, consumer protection, anti-money laundering, sanctions, outsourcing, operational resilience and the use of models. Rules differ across countries and can change as technology develops. A product launched quickly in one market may create licensing, disclosure or consumer-protection issues in another. Poorly designed digital interfaces can also lead to customer misunderstanding, unsuitable outcomes or allegations of unfair treatment. Compliance must be built into product design rather than added only after launch.

2.8 Reputational and trust risks

Customers may judge a bank not only by the security of its own systems but also by the behaviour of its partners and the fairness of its automated decisions. A prolonged outage, privacy incident, discriminatory algorithm or slow fraud response can damage confidence. Reputational harm may spread rapidly through social media and can lead to customer complaints, account closures, increased scrutiny and higher remediation costs. Transparent communication and timely support are essential during incidents.

3. How participation in FinTech markets can increase risk within HSBC

The risks above can arise from using technology internally. Participation in FinTech markets can add further exposure because the bank becomes more connected to external firms, new products, unfamiliar business models and rapidly changing customer expectations.

Investment and execution risk

Developing or acquiring technology, launching digital products and modernising legacy systems require capital, skilled staff and management attention. Projects may be delayed, exceed budgets or fail to deliver expected benefits. Running old and new platforms in parallel can temporarily increase complexity and operational risk.

Partnership and ecosystem risk

Collaborations with start-ups and technology providers can accelerate innovation, but partners may have different risk cultures, weaker control environments or limited financial resources. Integration failures, unclear accountability and incompatible security standards can create vulnerabilities. The bank should assess the full service chain, not only the direct contractual partner.

Competitive and strategic risk

Competition from digital banks, payment firms and other FinTech businesses may pressure HSBC to launch products faster, lower fees or change its operating model. Moving too slowly can result in lost customers and market share; moving too quickly can lead to inadequate testing, weak controls or poor customer outcomes. Strategic decisions must balance innovation speed with risk appetite.

New product and market risk

Entering digital assets, tokenisation, embedded finance or new payment services may expose a bank to uncertain legal treatment, market volatility, liquidity concerns, fraud and reputational consequences. Before entering a market, the firm needs to understand the product, customer demand, legal status, custody arrangements, settlement process and potential loss scenarios.

Interconnectedness and systemic exposure

FinTech markets link banks with payment networks, cloud providers, platforms and other financial firms. A failure can propagate through these connections. Common technology providers or shared infrastructure may create correlated exposures across institutions. This can turn a local operational incident into a broader stability concern.

Skills, culture and governance pressure

Competition for cybersecurity specialists, data scientists, software engineers and model-risk experts can increase costs and create capability gaps. Traditional governance processes may be too slow for agile product teams, while overly weak governance may allow risks to go unchallenged. Clear accountability, training and independent challenge are needed.

4. Risk interaction and potential consequences

FinTech risks rarely occur in isolation. For example, a compromised supplier account could provide access to a cloud-hosted application; the incident could expose customer data, interrupt payments, trigger regulatory reporting duties and damage the bank's reputation. Similarly, an inaccurate AI model could cause unfair customer decisions, generate complaints and lead to regulatory action. These examples show why HSBC should assess risks across people, processes, technology and external dependencies rather than treating each technology as a separate issue.

Risk source	Possible internal impact	Wider consequence
Cyber incident	Service disruption; investigation and recovery costs	Customer harm and reputational damage
AI/model failure	Incorrect decisions; manual rework	Conduct concerns and regulatory scrutiny
Vendor outage	Loss of access to critical services	Payment disruption and concentration exposure
Digital fraud	Direct losses; claims and case handling	Reduced trust and financial-crime exposure
Regulatory failure	Remediation and compliance costs	Penalties, restrictions or business changes

5. Conclusion – points to develop

FinTech can strengthen HSBC's customer experience, efficiency and ability to compete, but it also changes the bank's risk profile. Digital channels expand the attack surface; data-intensive services raise privacy and governance questions; AI creates model and conduct risks; and external partnerships increase dependency and concentration exposure. Participation in FinTech markets adds strategic, investment, integration and interconnectedness risks. The conclusion should emphasise that the appropriate response is not to avoid innovation, but to align innovation with a clearly defined risk appetite, strong governance, resilient systems, capable employees and effective oversight of partners.

6. Research and evidence checklist

- Read HSBC's latest Annual Report and Accounts and identify the sections on principal risks, technology, cyber security, operational resilience and financial crime.
- Check HSBC's official publications for current information on digital transformation, AI, data and technology risk. Do not rely on headlines alone.
- Use regulatory sources such as the Bank of England, Prudential Regulation Authority, Financial Conduct Authority, European Banking Authority and Financial Stability Board where relevant.
- For every company-specific statement, record the publication title, date, page number and URL.
- Add at least two concrete, properly sourced examples and explain how each example relates to the risk analysis. Do not imply that a hypothetical incident actually happened to HSBC.

Suggested sources to consult

HSBC – Annual reports and results

https://www.hsbc.com/investors/results-and-announcements

HSBC – Risk management

https://www.hsbc.com/who-we-are/esg-and-responsible-business

Bank of England – Operational resilience

https://www.bankofengland.co.uk/operational-resilience

Financial Stability Board – FinTech and financial stability

https://www.fsb.org/work-of-the-fsb/financial-innovation-and-structural-change/

Financial Action Task Force – Digital transformation and financial crime resources

https://www.fatf-gafi.org/

Before submitting

This guide does not replace source verification or your own analysis. Add your course concepts, explain why each risk is material to HSBC, cite the exact evidence you use, and make sure the final report meets your instructor's word-count and formatting requirements. The Coursera assignment page states that AI-generated submissions may be treated as an Honor Code violation.